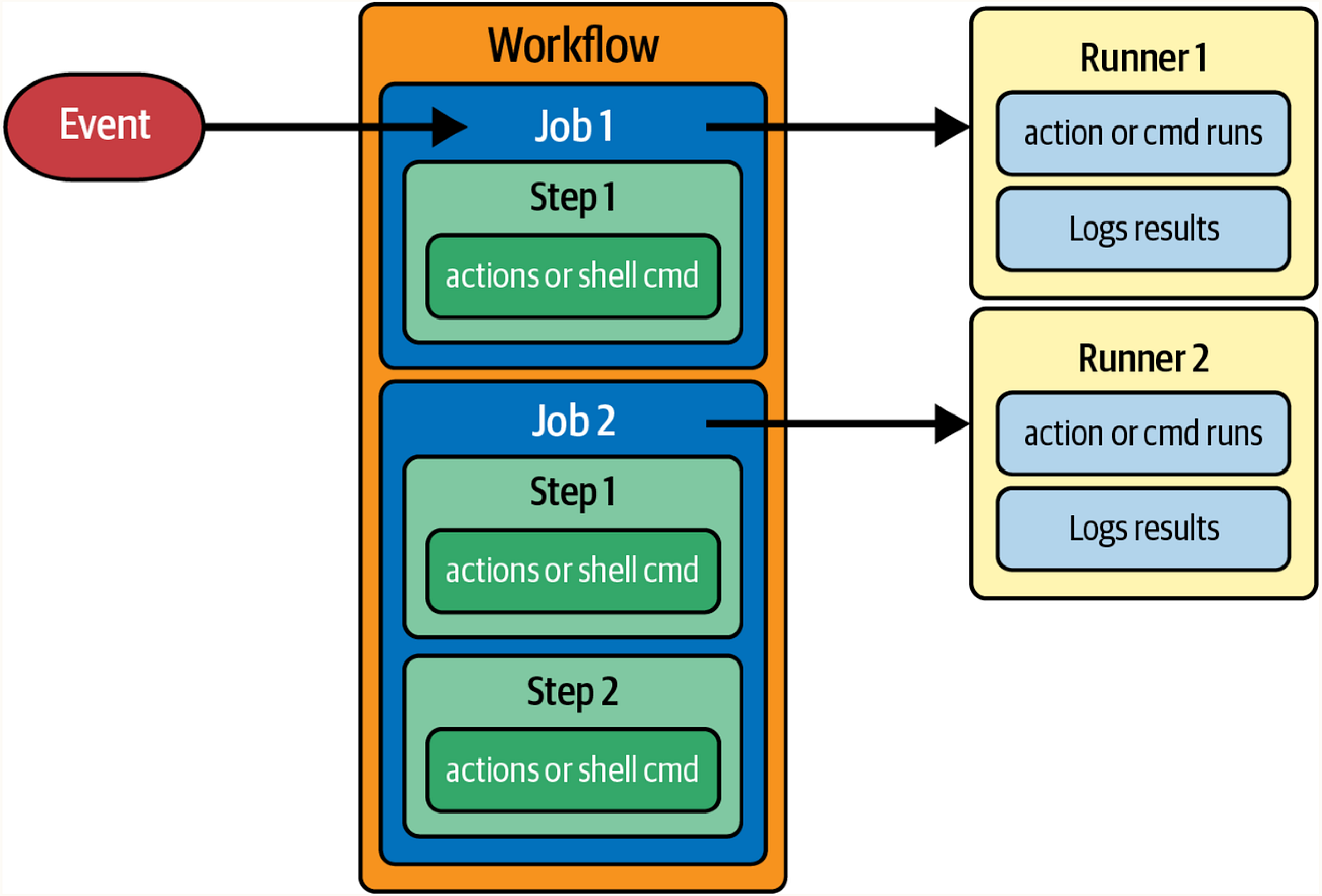
Lab 6: More on GitHub Actions

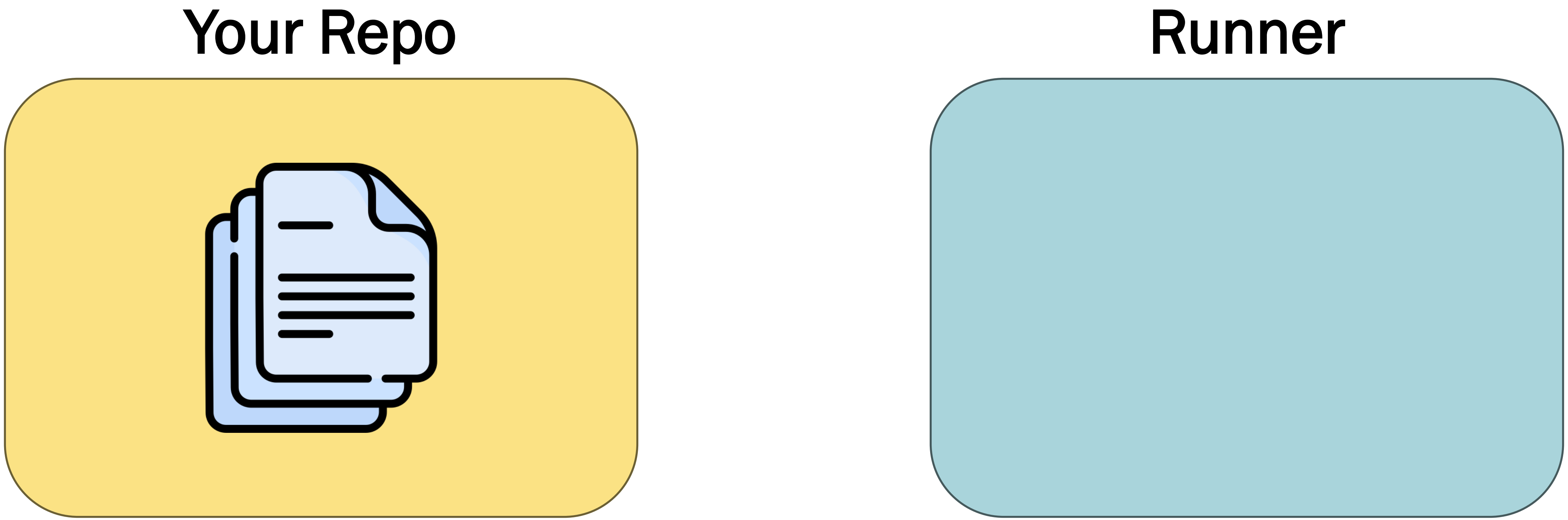
CSE233 Agile Software Engineering


Event
Workflow
Job 1
Step 1
actions or shell cmd
Job 2
Step 1
actions or shell cmd
Step 2
actions or shell cmd
Runner 1
action or cmd runs
Logs results
Runner 2
action or cmd runs
Logs results

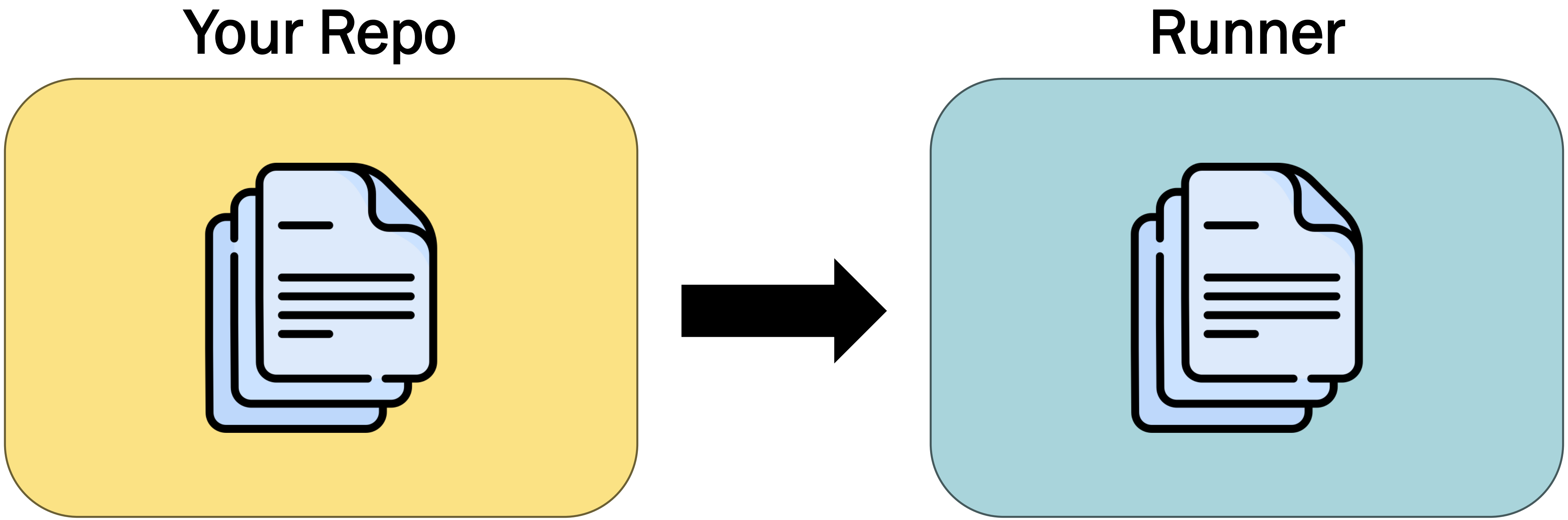
actions/checkout

- This is usually the **first step** in most workflows.
- It pulls your repository’s code into the runner.
- Without checkout, your runner won’t have the repo files.

```yaml
steps:
  - name: Checkout code
    uses: actions/checkout@v4
```




```yaml
steps:
  - name: Checkout code
    uses: actions/checkout@v4
```




```yaml
steps:
  - name: Checkout code
    uses: actions/checkout@v4
```

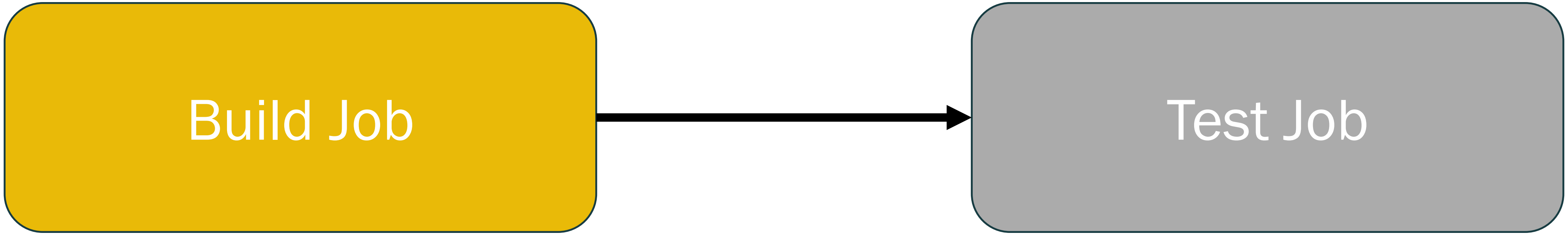
Job Dependencies

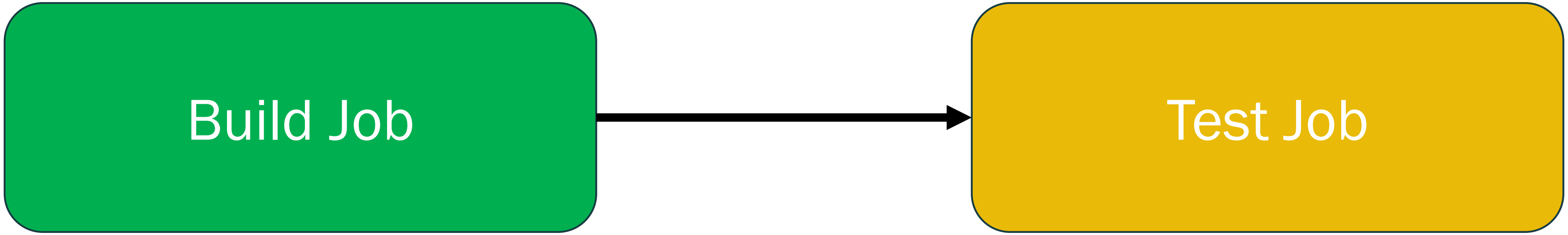
- **needs**: defines which jobs must succeed before running the next one.
- Enables sequential pipelines: *build* → *test* → *deploy*
- Prevents unnecessary execution if earlier stages fail

```yaml
jobs:
  build:
    runs-on: ubuntu-latest
    steps:
      - run: echo "Building project..."

  test:
    runs-on: ubuntu-latest
    needs: build
    steps:
      - run: echo "Testing after build!"
```

- **Build** runs first → **Test** waits until it succeeds.


Build Job
Test Job


Build Job
Test Job

Conditional Logic

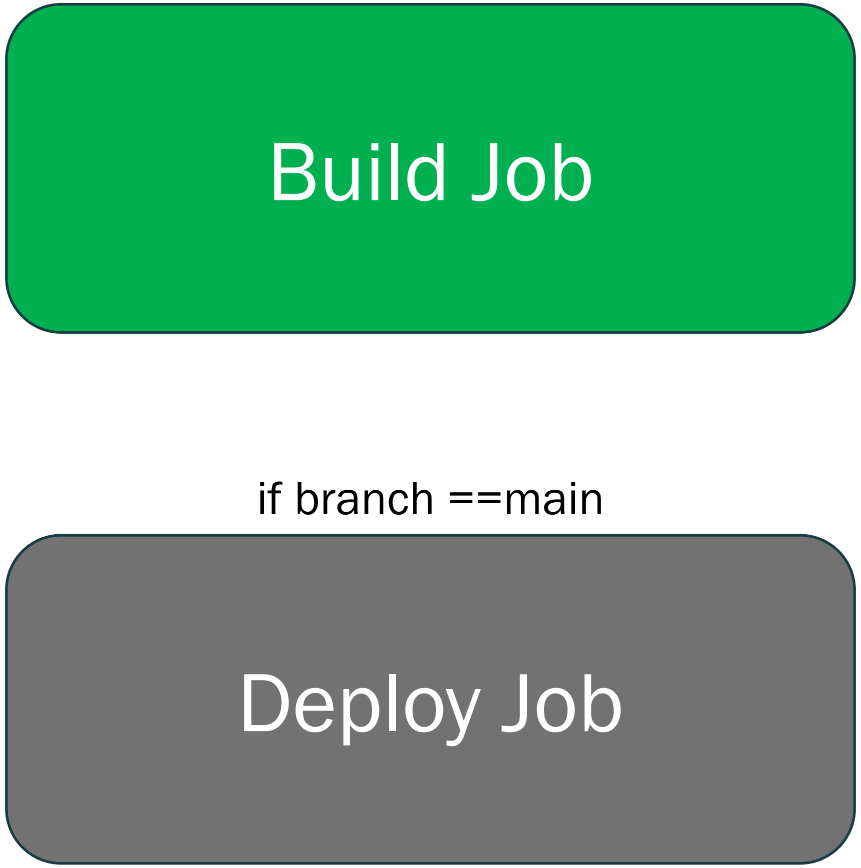
- **if**: controls when a job or step should run.
- Useful for:
  - Deploy only from the main branch
  - Skip certain tests
  - Notify only on failure

```
if: github.ref == 'refs/heads/main'
```

Conditional Logic

- **if**: controls when a job or step should run.
- Useful for:
  - Deploy only from the main branch
  - Skip certain tests
  - Notify only on failure

```
if: github.ref == 'refs/heads/main'
```


Build Job
if branch ==main
Deploy Job

Matrix Builds

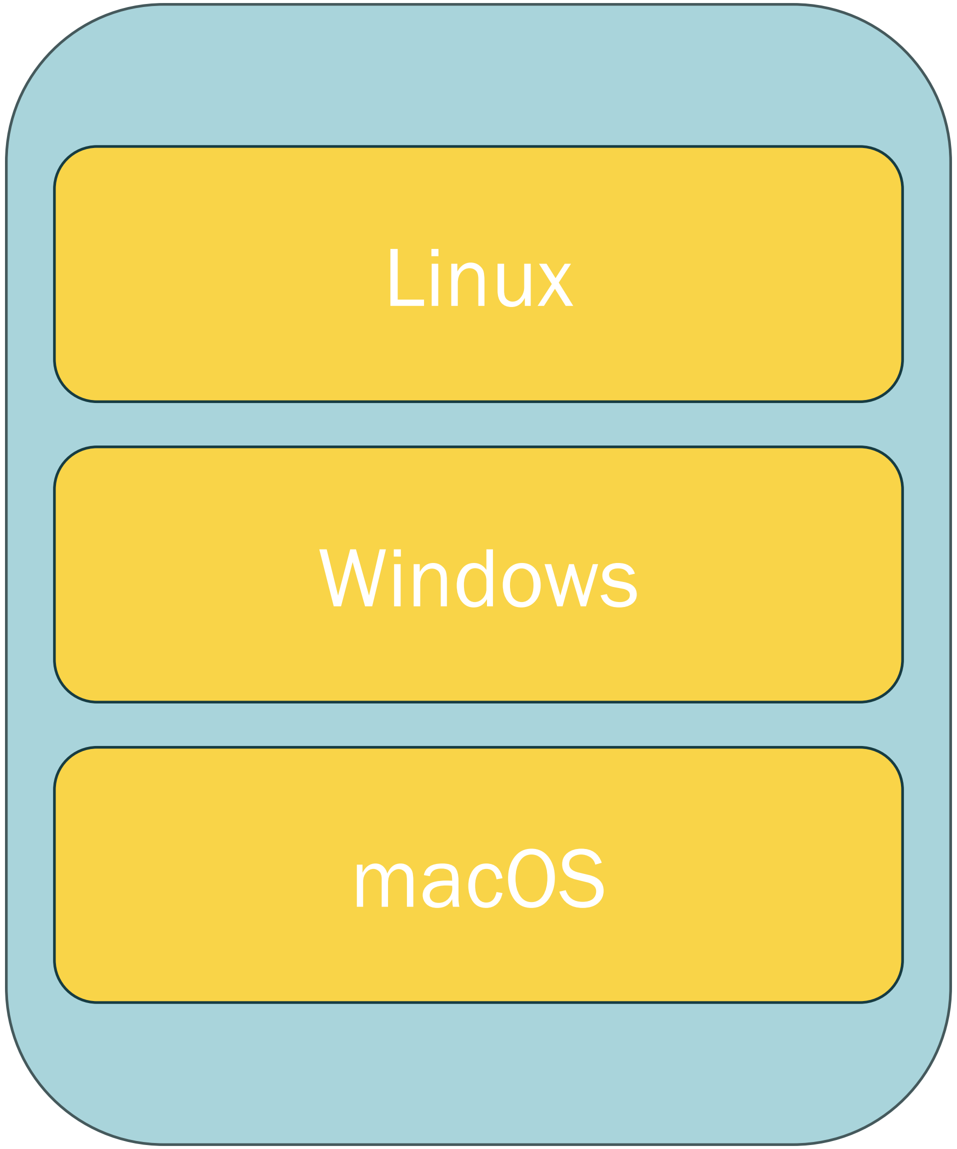
- A **matrix** allows one job to run on multiple configurations simultaneously (e.g., OS, Python versions).
- Saves time by testing all environments in one workflow

```yaml
strategy:
  matrix:
    os: [ubuntu-latest, windows-latest, macos-latest]
runs-on: ${{ matrix.os }}
steps:
  - run: echo "Running on ${{ matrix.os }}"
```

Linux

Windows

macOS


Linux
Windows
macOS

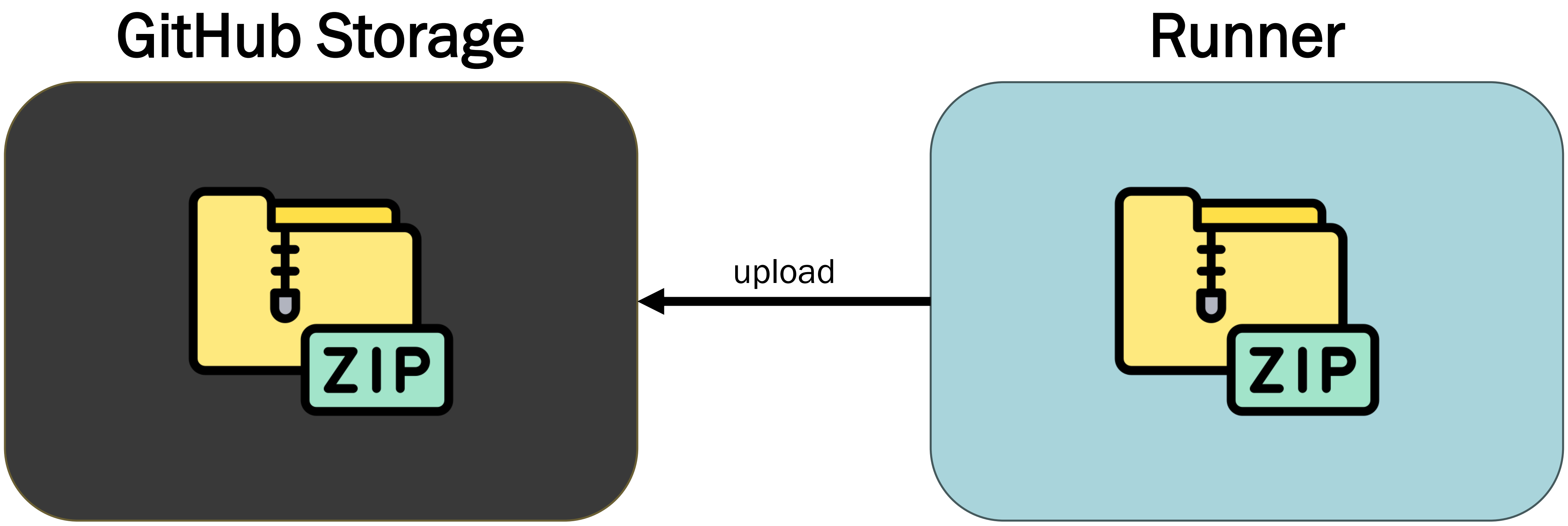
Artifacts

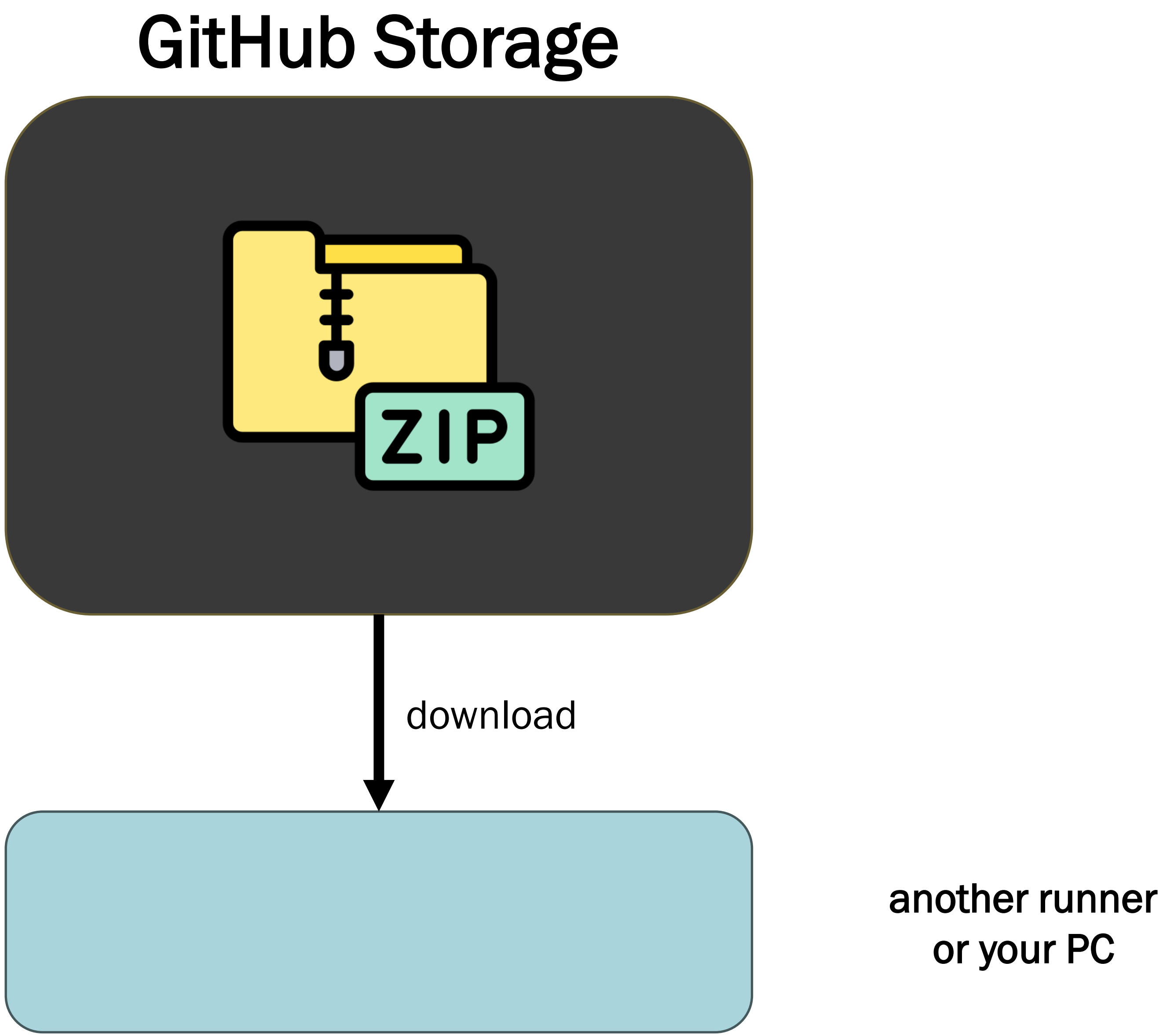
- Each job runs on a clean runner → no shared state.
- To pass files between jobs, use **artifacts**.

```yaml
- name: Upload build results
  uses: actions/upload-artifact@v4
  with:
    name: build-output
    path: build/
```

```yaml
- name: Download build results
  uses: actions/download-artifact@v4
  with:
    name: build-output
```

Runner


GitHub Storage
Runner
upload
ZIP
ZIP


GitHub Storage
ZIP
download
another runner
or your PC

Environment Variables

- **Environment variables** store reusable values across jobs or steps.
- Useful for:
  - Setting file paths, names, tokens, constants
  - Avoiding repetition

```yaml
env:
  PROJECT_NAME: my-app
  VERSION: 1.0

jobs:
  build:
    runs-on: ubuntu-latest
    steps:
      - run: echo "Building $PROJECT_NAME version $VERSION"
```

Passing Data Between Steps

- You can export variables dynamically using $GITHUB_ENV.
- Helps share runtime-generated data between steps.

```yaml
- name: Generate random ID
  run: echo "RUN_ID=$(date +%s)" >> $GITHUB_ENV

- name: Use it later
  run: echo "Run ID is $RUN_ID"
```

Secrets

- Store sensitive info safely (API keys, tokens) as **Secrets.**
- Protects credentials from exposure

```
env:
  DEPLOY_KEY: ${{ secrets.DEPLOY_KEY }}
```

Thank you!

Eng. Abdelrahman Salah